

Aurora Gaeta

Curriculum Vitae

July 20, 2026

Email II: auroragaeta@gmail.com
Languages: Italian (native), English (fluent)
Citizenship: Italian

Profile

Biostatistician and PhD Candidate with expertise in real-world data analysis, epidemiology, and predictive modelling in healthcare. Experience in translating complex clinical and epidemiological data into actionable insights to support decision-making in oncology and public health. Strong background in multidisciplinary collaboration and international research projects involving real-world evidence generation.

Research Experience

Aug 2021 – Present **Research Fellow in Biostatistics (Real-World Data & Evidence)**
European Institute of Oncology, Milan

- Contributed to research projects funded by the IEO-MONZINO Foundation and the Italian Ministry of Health (PNRR).
- Conduct independent and collaborative research in oncology, epidemiology, and AI applications.
- Developed predictive models using real-world clinical and epidemiological data to support risk stratification and clinical decision-making.
- Led data analyses on multicentric real-world datasets, translating results into insights supporting clinical research and evidence-based decisions.
- Contribute to study design, statistical analysis plans, and ethics submissions.
- Co-author of peer-reviewed publications and international research outputs.

Mar 2023 – Present **Artificial intelligence and Radiomic board scientific secretariat**
European Institute of Oncology, Milan

Sep 2025 – Feb 2026 **Visiting Research Scholar**
Harvard Medical School – Brigham and Women's Hospital, USA

- Developed pseudo-value based methodology for survival outcomes and model evaluation in precision medicine.
- Conduct statistical analyses in cardiovascular clinical research (TIMI Study Group).
- Collaborate with international teams in biostatistics and epidemiology.

Core Competencies

- **Real-World Evidence and Clinical Data Analysis:** analysis of real-world oncology datasets, patient-level clinical data, treatment outcomes, toxicity profiles, and prognostic factors.
- **Advanced Statistical Methods:** survival analysis, competing risks, multivariable regression, meta-analysis, systematic reviews, predictive modelling, validation studies.

- **Oncology and Epidemiology:** melanoma, skin cancer, breast cancer, head and neck cancer, lung cancer, prostate cancer, gynecological cancer, cancer prevention and survivorship.
- **AI in Healthcare:** Proteomics, Radiomics and AI in Healthcare: analysis of Olink proteomic data for cardiometabolic and kidney risk biomarker discovery; analysis of imaging-derived features from PET, CT, MRI, ultrasound, contrast-enhanced mammography; AI applications in cancer screening and outcome prediction.
- **Evidence Generation:** study design, statistical analysis plans, protocol development, literature synthesis, interpretation of results, scientific reporting.
- **Programming and Data Management:** R, Python, SQL, data cleaning, statistical reporting, reproducible analysis workflows.

International Research Collaborations

- **Euromelanoma Program**
Contribution to epidemiological research, prevention strategies, and analysis of behavioural and clinical data in melanoma. <https://www.euromelanoma.eu/en-int1/>
- **M-SKIP Network (International Multicentric Collaboration)**
Development of predictive models and statistical analysis of clinical, epidemiological, and genetic data in skin cancer research.
- **Artificial Intelligence and Radiomics Board – IEO**
Methodological and statistical evaluation of AI- and radiomics-based oncology projects. <https://www.ieo.it/en/RESEARCH/Radiomic/>
- **AIRC-funded Research Project**
Radiogenomics for predicting underestimation of invasiveness in DCIS diagnosed with Vacuum Assisted Breast Biopsy” (AIRC GI 2021 - DI 25816)

Applied ML and statistical modelling to high-dimensional radiomics data for cancer risk stratification, integrating clinical and imaging features to construct predictive signatures, with rigorous cross-validation and external cohort validation.

Education

- 2023 – [2026] **PhD in Public Health, Epidemiology, Statistics and Economics**
University of Milano-Bicocca, ends October 2026, discussion March 2027
- 2020 – 2023 **MSc in Biostatistics (110/110 cum laude)**
University of Milano-Bicocca
- 2017 – 2020 **BSc in Economics and Statistics**
University of Torino

Honors and Research Funding

- Competitive research fellowships (IEO-MONZINO Foundation).
- Fellowship renewal based on performance.
- PhD funded within the Italian National Recovery and Resilience Plan (PNRR – Ministry of Health) and by IEO-MONZINO Foundation

Teaching Experience

- 2024 – 2025 **Adjunct Lecturer – SQL and Data Management**
Master in Data Management for Clinical Research
- 2025 **Teaching Assistant – AI and Data Analytics (AIDA Master)**
- 2023 – 2025 **Teaching Assistant – Machine Learning (MSc Biostatistics)**
- 2023 – 2024 **Tutor – Gender Equality Plan (GEP)**

Technical Skills

- Statistical modelling, survival analysis, competing risks, machine learning.
- Study design, statistical analysis plans, and clinical research methodology.
- Programming: R, Python, SQL.

Selected Presentations

- Oral presentation International Biometric Conference, Seoul, 2026, 'Assessing calibration in regression and ML-based prediction models using pseudo-values for time-to-event outcomes'
- Oral presentation, International Biometric Conference, USA, 2024, "M-SKIP Project: Melanoma risk and mole count across age groups".
- Oral presentation, International Society of Gender Medicine, Italy, 2022, "The prognostic role of sex and anemia in tongue cancer patients".

Selected Poster

- **IARC@60** International Conference 2026 (International Agency for Research on Cancer): Gaeta, A., Pierini, L.,... S., & Gandini, S.
"Effectiveness of skin cancer prevention interventions in melanoma survivors: A systematic review."
- **IARC@60** International Conference 2026 (International Agency for Research on Cancer): Gaeta, A., Njimi, H., Bulliard, J. L.,..., Del Marmol, V., Gandini, S., & Suppa, M.
"Indoor tanning: Prevalence, trends and determinants from the Euromelanoma Campaign."
- **AHA EPI** —Lifestyle Scientific Sessions (Epidemiology and Prevention — Lifestyle and Cardiometabolic Health, American Heart Association) 2026:
"Assessing Calibration and Recalibration of Clinical Prediction Models Using Pseudo-Values Regression" Andrea Bellavia , PhD , Aurora Gaeta , MS , e Sabina Murphy , MPH
https://www.ahajournals.org/doi/10.1161/cir.153.suppl_1.TH875
- **ASCO** (American Society of Clinical Oncology): Ferrucci, P. F., Gaeta, A., Cocorocchio, E., D'Ecclesiis, O., Gandini, S. (2022).
"Meta-analysis of randomized phase II-III trials evaluating triplet combinations of immunotherapy and targeted therapy for BRAF V600- mutant unresectable or metastatic melanoma."
- **ASCO** : Valenza, C., Olmeda, E., Porta, F. M., De Pas, T. M., Gaeta, A., Castellano, G., ... Curigliano, G. (2024).
"The prevalence of fibroadenoma-like areas in malignant phyllodes tumors of the breast: A retrospective analysis from a referral center."
- **Annals Of Oncology** Valenza, C., De Pas, T. M., Castellano, G., Santoro, C., Corona, A., Chiappini, G., ... Curigliano, G. (2023)
"Non-metastatic malignant phyllodes tumors of the breast (B-MPT): A retrospective analysis from a referral center. Annals of Oncology, 34, S1042- S1043."
- **IGM** (international society of gender medicine 2022)
"Are Sex and Gender Considered in Head and Neck Cancer Clinical Study Designs?" Gaeta et.al
- **ACC** (Alleanza contro il cancro) "Reporting melanoma in Covid-era". Aurora Gaeta et.al. (2022)
- **ACC** "Women's perceptions and attitudes to the use of AI in breast cancer screening: a survey in a cancer referral center." Pesapane, et.al.(2022)
- **ACC** "Meta-analysis of randomized phase II-III trials evaluating triplet combinations of immunotherapy and targeted therapy for BRAF V600-mutant unresectable or metastatic melanoma"
" Ferrucci, Gaeta, D'Ecclesiis, Gandini.(2022)

Scientific Output

- 47 peer-reviewed publications.
- H-index: 10.
- ORCID <https://orcid.org/0000-0002-4944-7685>

Scientific Conferences and Training

- IARC@60 International Conference, 19 – 21 May 2026
- 32nd International Biometric Conference 8-13 December 2024 Atlanta, USA
- StaTalk , University of Milano-Bicocca, 13-14 June 2025
- Alliance Against Cancer Annual Meeting, Rome, 21-23 September 2022
- IGM (international society of gender medicine), 16–17 2022 September

Organization of science-related events

- Early Career Biostatisticians Subcommittee member (ISCB ECB) <https://iscb.international/early-career-biostatisticians/>
- Easy PhD Talks: Sharing to Improve Public Health. Seminar Series for PhD students in Public Health Epidemiology, Statistics and Economics.
- Organizing the grand round of Artificial Intelligence and Radiomics at the European Institute of Oncology

Selected Publications

- [1] Doi*, G., Gaeta*, A. c., Ribero, S., Gruis, N., Newton-Bishop, J., Polsky, D., Lazovich, D., Ghiorzo, P., Ribas, G., Menin, C., Stratigos, A. J., Guida, G., Puig, S., Fargnoli, M. C., Kanetsky, P. A., Queirolo, P., Bagnardi, V., Bataille, V., Raimondi, S., and Gandini, S. (2026). Age-specific melanoma risk associated with nevi: a pooled analysis from the m-skip project. *JDDG - Journal of the German Society of Dermatology*. Cited by: 0; All Open Access, Hybrid Gold Open Access.
- [2] Gaeta*, A., Nuvoli*, L., Doccioli, C., Caini, S., Saponara, M., Cimminiello, C., Cosma, C., Palmieri, G., Cossu, A., Vicini, F., Mazzearella, L., Tosti, G., Queirolo, P., and Gandini, S. (2026). Sex and gender influence on adverse events for melanoma patients: a comprehensive review and meta-analysis. *British Journal of Cancer*, 134(3):359 – 366. Cited by: 0; All Open Access, Green Open Access.
- [3] Gaeta*, A., Tagliabue*, M., D'Ecclesiis, O., Ghiani, L., Maugeri, P., De Berardinis, R., Veneri, C., Gaiaschi, C., Cacace, M., D'Andrea, L., Ansarin, M., Gandini, S., and Chiocca, S. (2023). Are sex and gender considered in head and neck cancer clinical studies? *npj Precision Oncology*, 7(1). Cited by: 14; All Open Access, Gold Open Access, Green Open Access.
- [4] Marvaso, G., Jerezek-Fossa, B. A., Zaffaroni, M., Vincini, M. G., Corrao, G., Andratschke, N., Balagamwala, E. H., Bedke, J., Blanck, O., Capitanio, U., Correa, R. J. M., De Meerleer, G., Franzese, C., Gaeta, A., Gandini, S., Garibaldi, C., Gerszten, P. C., Gillessen, S., Grubb, W. R., Guckenberger, M., Hannan, R., Jhaveri, P. M., Josipovic, M., Kerkmeijer, L. G. W., Lehrer, E. J., Lindsog, M., Louie, A. V., Nguyen, Q.-N., Ost, P., Palma, D. A., Procopio, G., Rossi, M., Staehler, M., Tree, A. C., Tsang, Y. M., Van As, N., Zaorsky, N. G., Zilli, T., Pasquier, D., and Siva, S. (2024). Delphi consensus on stereotactic ablative radiotherapy for oligometastatic and oligoprogressive renal cell carcinoma—a european society for radiotherapy and oncology study endorsed by the european association of urology. *The Lancet Oncology*, 25(5):e193 – e204. Cited by: 31.
- [5] Sattin, C., Pizzi, C., Summers, P., Gaeta, A., Gandini, S., Alessi, S., Nguyen, H. Q., Renne, G., Luzzago, S., Marvaso, G., Musi, G., Jerezek-Fossa, B. A., Padhani, A. R., and Petralia, G. (2025). Histopathologic yields and concordance of in-bore mri-targeted biopsy for prostate cancer diagnosis. *Radiology*, 316(1). Cited by: 2.